

# JUSTIN PAYAN

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[LinkedIn](#) ♦ [Github](#) ♦ [Google Scholar](#)

## ACADEMIC POSITIONS

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**Carnegie Mellon University**

Oct '24 - Present

*Postdoctoral Research Associate with [Nihar Shah](#)*

Pittsburgh, PA

- Applying large language models to improving scientific reasoning and peer review
- Developing dynamic scheduling algorithms for ground-based telescopes

## RESEARCH INTERESTS

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I optimize scientific productivity using techniques from operations research, data science, machine learning, and natural language processing.

## EDUCATION

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**University of Massachusetts Amherst**

Sept '18 - February '25

PhD Computer Science

(advised by [Yair Zick](#))

**University of Georgia**

Sept '13 - May '18

M.S. Artificial Intelligence, B.S. Mathematics, A.B. Cognitive Science

(advised by [Frederick Maier](#))

## INDUSTRY EXPERIENCE

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**Adobe Research**, Document Intelligence Lab

May - August '23

*Research Intern under [Chris Tensmeyer](#)*

College Park, MD

- Document structure inference using deep learning, computer vision, and combinatorial optimization

**Microsoft**, Calc Intelligence

June - September '22

*Data Science Intern under [Kartik Sridhar](#)*

Redmond, WA

- Data augmentation using GPT-3 for an NL to code task
- Worked closely with both production and research teams to implement state-of-the-art modeling and evaluation

**Amazon Alexa**, Trustworthy Alexa

June - September '21

*Research Intern under [Yuval Merhav](#)*

Cambridge, MA (remote)

- Applied generative insertion transformers (GIT) to data augmentation for NER
- Investigated interaction of GIT augmentation with knowledge-based augmentation approaches

**Amazon Alexa**, Trustworthy Alexa

May - August '20

*Research Intern under [Yuval Merhav](#)*

Cambridge, MA (remote)

- Explored the efficacy of generative replay for continual learning in a privacy-aware setting

- Created and released dataset for evaluating continual learning for single-task NER

**Vertica**, Machine Learning

June '17 - June '18

*Software Engineer under Vincent Xu*

Cambridge, MA

- Implemented distributed machine learning algorithms in SQL and C++, including k-means++
- Designed, built, and maintained data preprocessing functions, such as one-hot encoding, normalization, and missing value imputation

## PUBLICATIONS

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$(\alpha - \beta)$  indicates alphabetical order; asterisks indicate equal contribution.

1. [ML Researchers Support Openness in Peer Review But Are Concerned About Resubmission Bias](#)  
Vishisht Rao, **Justin Payan**, Andrew McCallum, and Nihar B. Shah  
**Under Submission**
2. [FLAWS: A Benchmark for Error Identification and Localization in Scientific Papers](#)  
Sarina Xi, Vishisht Rao, **Justin Payan**, and Nihar B. Shah  
**Under Submission**
3. Uncertainty-aware Ground-based Telescope Observation Scheduling at ALMA  
**Justin Payan**, Noemi Barbagli, Ignacio Toledo, Rodrigo A. Carrasco, Sergio Martín, and Nihar B. Shah  
**SPIE Astronomical Telescopes and Instrumentation 2026**
4. [Fair and Welfare-Efficient Constrained Multi-matchings under Uncertainty](#)  
Elita Lobo\*, **Justin Payan\***, Cyrus Cousins, and Yair Zick  
**NeurIPS 2024**
5. [InstructExcel: A Benchmark for Natural Language Instruction in Excel](#)  
**Justin Payan**, Swaroop Mishra, Mukul Singh, Carina Negreanu, Christian Poelitz, Chitta Baral, Subhro Roy, Rasika Chakravarthy, Benjamin Van Durme, and Elnaz Nouri  
**EMNLP 2023 (Findings)**
6. [Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities](#)  
Cyrus Cousins, **Justin Payan**, and Yair Zick ( $\alpha - \beta$ )  
**SAGT 2023**
7. [Graphical House Allocation](#)  
Hadi Hosseini, **Justin Payan**, Rik Sengupta, Rohit Vaish, and Vignesh Viswanathan ( $\alpha - \beta$ )  
**AAMAS 2023**
8. [Relaxations of Envy-Freeness over Graphs](#)  
**Justin Payan**, Rik Sengupta, and Vignesh Viswanathan ( $\alpha - \beta$ )  
**AAMAS 2023 (Extended Abstract)**
9. [I Will Have Order! Optimizing Orders for Fair Reviewer Assignment](#)  
**Justin Payan** and Yair Zick  
**IJCAI 2022**

10. [Towards Realistic Single-Task Continuous Learning Research for NER](#)  
**Justin Payan**, Yuval Merhav, He Xie, Satyapriya Krishna, Anil Ramakrishna, Mukund Sridhar,  
and Rahul Gupta  
**EMNLP 2021 (Findings)**
11. [Online Post-Processing in Rankings for Fair Utility Maximization](#)  
Ananya Gupta\*, Eric Johnson\*, **Justin Payan**, Aditya Roy, Ari Kobren, Swetasudha Panda,  
Michael Wick, and Jean-Baptiste Tristan  
**WSDM 2021**
12. [Document Representations using Fine-Grained Topics](#)  
**Justin Payan**, Nicholas Monath, and Andrew McCallum  
**Sets & Partitions Workshop at NeurIPS 2019**

## CODE CONTRIBUTIONS

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### OpenReview

2022

Created and deployed the [FairSequence reviewer matching algorithm](#) on OpenReview platform, used by over 50 peer review venues

## AWARDS & SCHOLARSHIPS

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- Google Cloud Computing Credits (Spring 2026)  
Amount: Equivalent of \$30,000
- UMass CICS Proposal Writing Grant (Spring 2023)  
Amount: \$12,733.80
- University of Georgia Foundation Fellowship (2013-2017)  
\$128,260 in tuition and stipend for housing, research, and travel

## INVITED TALKS

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- Brown University (E-GLAMOR Group), March 2023  
Harvard University (EconCS Seminar), January 2023  
Carnegie Mellon University (Nihar Shah's Group), November 2022  
(*Into the Unknown: Assigning Reviewers to Papers with Uncertain Affinities*)
- UMass Data Analytics and Computational Social Science Brownbag Series, April 2021  
(*Envy-Freeness in Paper Reviewer Assignment*)
- UMass Amherst Theory Seminar, April 2021  
(*Fair Reviewer Assignment*)

## ACADEMIC SERVICE

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- **Workflow Chair** for IJCAI 2023
- **Workshop Co-organizer** for [Computational Fair Division](#) at IJCAI 2023-2025, [Practical Improvements to Peer Review](#) at AAMAS 2025, [Workshop on Incentives in Academia](#) at EC 2025
- **Program Committee / Reviewer** for ACL 2023, EC 2025, KDD 2025

- **Subreviewer** for NeurIPS 2024, EC 2024, AAMAS 2022, GAIW at AAMAS 2022

## TEACHING / MENTORING / OTHER SERVICE

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- **Instructor of Record** at UMass Amherst for *Intro to Numerical Computing with Python* (Fall 2020 & 2021)
- **Graduate Teaching Assistant** at UMass Amherst for *Intro to AI* (Fall 2018), *Intro to Problem Solving with Computers* (Fall 2020), and *Advanced Algorithms* (Spring 2021)
- **Mentees:**
  - Rudra Gopal. BS, University of Massachusetts. Now MS at Georgia Tech.
  - Gautam Chatterjee. BS, University of Massachusetts. Now Software Engineer at Wayfair.
  - Ed Almusalamy. BS, University of Massachusetts. Now MS at University of Massachusetts.
  - Hassan Gudal. BS, University of Massachusetts. Graduated 2024.
  - Noemi Barbagli. BS, Carnegie Mellon University.
  - Eric Johnson. MS, University of Massachusetts. Graduated 2021.
  - Ananya Gupta. MS, University of Massachusetts. Now MLE at Diffbot.
  - Aditya Kumar Roy. MS, University of Massachusetts. Graduated 2021.
  - Vedant Bhatia. MS, University of Massachusetts. Now Quantitative Developer at HBK Capital Management.
  - Vishisht Rao. MS, Carnegie Mellon University.
  - Sarina Xi. MS, Carnegie Mellon University.
  - Arshika Lalan. MS, Carnegie Mellon University.

## SKILLS

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Python, Gurobi, CVXPY, linear programming, integer programming, combinatorial optimization, machine learning, named entity recognition, large language models, Pandas, Numpy, Scikit-learn, Pytorch